

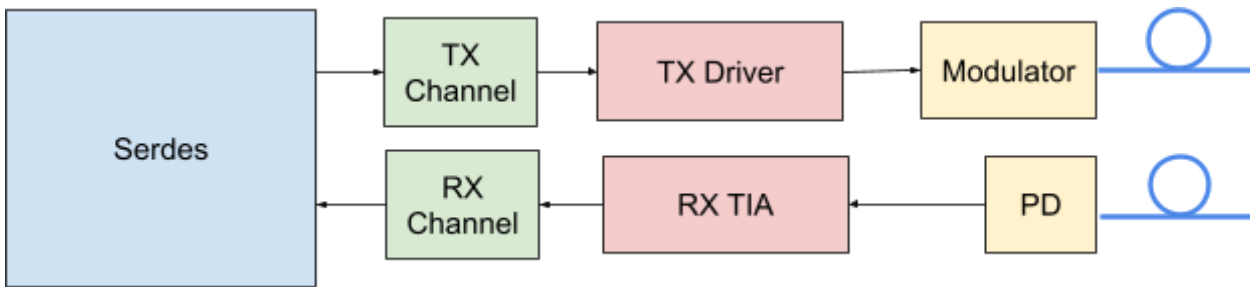
Requirements Specifications for the IP cores, 224G RX TIA & TX DRIVER and 64G RX TIA & TX DRIVER blocks

Revision	Date	Author	Description
0.1	11/16/2025	Nicolas Monier Hasan Al-Rubaye Cathy Huang	First Draft Created
0.2	11/25/2025	Hasan Al-Rubaye Cathy Huang	Electrical requirements based on architecture document Rev. 0.25
0.3	12/1/2025	Hasan Al-Rubaye	Clarified TIA gain requirements
0.4	12/3/2025	Hasan Al-Rubaye	Added DCOC loop specs
0.5	12/5/2025	Nicolas Monier	Added specs for range and rms noise of DC current output Added Project Timeline and Deliverables
0.6	12/12/2025	Nicolas Monier	Added X/Y dimensions for each IP
0.7	01/06/2026	Nicolas Monier	Renamed document "Requirements Specifications for the IP cores, 224G RX TIA & TX DRIVER and 64G RX TIA & TX DRIVER blocks" Added Process, Voltage, and Temperature corners, sign-off requirements
0.8	01/07/2026	OmniDesign	Omni comments against requirements
0.9	01/07/2026	Nicolas Monier	Response to Omni comments Removed Project timeline, which is now in the SOW
0.10	01/09/2026	Cathy Huang	Added paragraph about Output Bias Circuit - spec TBD

0.11	01/15/2026	Cathy Huang Clifford Chao Alexander Sludds Nicolas Monier	Updated specifications to match the latest architecture requirements Added new comments to add clarity on specifications and conditions
0.12	05/12/2026	Nicolas Monier	Changed the gain of the 224G TIA from 72-58 dB to 66-54 dB

This document outlines the requirements for the custom analog IPs Lightmatter is seeking in the TSMC N3P process node.

These circuits include the TX Driver and the RX TIA, which serve as the primary building blocks interfacing with the silicon-photonics elements. Lightmatter is looking for different designs to support multiple data rates and encoding schemes: 224 Gbps PAM4 and 56-64 Gbps NRZ.



TX Driver Circuits

224G PAM4 Linear Driver

Operational specification

Metric	Requirement			Unit	Comments
	Min	Typ	Max		
Data operating range (PAM4)	106.25		224	Gbps	
Power Consumption			0.7	pJ/bit	This power consumption requirement applies to an output impedance of 60 Ohms and needs to be revisited by OmniDesign

					and Lightmatter to achieve the new 30 Ohms requirements Power may be traded off to achieve schedule and other key specs (output swing) but should not exceed 0.9 pJ/bit
Differential Input Resistance		100		Ohms	
Differential Input Swing, Peak-to-Peak	0.5		1	Vpp	
Diff Output Swing, Peak-to-Peak	3			Vpp	TSMC has confirmed the max deltaV across any 2 nets in the N3P chip is 3.3V (with +/-10%, so 3.63V)!
Mid-Band Gain			16	dB	This is the MAX gain. The Gain control range lets it go down to 10db: - going from serdes max swing 1V into driver, to 3V out to MRM would want ~10db -we want the knobs to be able to go more than 10db since the serdes may not always be able to hit 1V swing (eg if we are using FFE and peaking).
Gain control range		6		dB	Programmable in 1dB steps (down to 10dB). Preserves frequency response. Verification coverage may be relaxed to achieve schedule, with full gain-range verification at pre-layout, and post-layout verification at all gain settings at TT and at the minimum (10 dB) and maximum (16 dB) gain settings across

					SS and FF corners.
In-band Gain Ripple			0.3	dB	
Equalization: Gain Peaking			4	dB	This is de-emphasis, meaning that in max gain condition we have 12dB mid-band gain.
Equalization: Gain Peaking Step		0.5		dB	
High-pass 3dB Bandwidth			100	kHz	
Low-pass 3dB Bandwidth	60			GHz	Defined as frequency at which the gain is 3dB below mid-band gain under all conditions BW may be relaxed to achieve other key specs (such as output swing) - subject to Lightmatter approval and alignment on schedule impact.
Output Differential DC Impedance			30	Ohms	Changed name from Differential Resistive Output Load to Output Differential DC Impedance
Differential Capacitive Output Load			60	fF	
THD			3	%	
Group Delay Variation			3	ps	Horizontal eye opening
X-Dimension			0.40	mm	Tentative, to be confirmed
Y-Dimension			0.20	mm	Tentative, to be confirmed
Poly orientation		Y			
Edge with high-speed signals		X edge			Tentative, to be confirmed

56-64G NRZ Driver

Electrical Requirements

Metric	Requirement			Unit	Comments
	Min	Typ	Max		
Data operating range (NRZ)	28		64	Gbps	
Power Consumption			0.3	pJ/bit	Power may be traded off to achieve key specs and schedule, but should not exceed 0.4 pJ/bit
Different Input Resistance	80	100	120	Ohms	
Differential Input Swing, Peak-to-Peak	0.9		1.2	Vpp	
Diff Output Swing, Peak-to-Peak	3			Vpp	Please report rms output noise
Diff Output Electrical Eye Width	12			ps	Under SSPR-NRZ data pattern @ 1e-12
High-pass 3dB Bandwidth			50	kHz	
Low-pass 3dB Bandwidth			32	GHz	Optional (use as guideline). Eye mask is more important
Capacitive Output Load			60	fF	
X-Dimension			0.3	mm	Tentative, to be confirmed
Y-Dimension			0.15	mm	Tentative, to be confirmed
Poly orientation		Y			
Edge with high-speed signals		X edge			Tentative, to be confirmed

Output Bias Circuit

At the output of DRV, an output bias circuit is used to set the voltage bias and couple the data signal to each MRM, provide differential termination, and reduce high-frequency common-mode noise and interference. The details spec of this bias circuit is TBD.

RX TIA (Transimpedance Amplifier) Circuits

224G PAM4 TIA

Electrical Requirements

Metric	Requirement			Unit	Comments
	Min	Typ	Max		
Data operating range (PAM4)	106.25		224	Gbps	
Power Consumption			0.2	pJ/bit	Power may be traded off to achieve key specs and schedule, but should not exceed 0.3 pJ/bit
Max Transimpedance Gain	66			dBΩ	Mid-band gain
Min. Transimpedance Gain			54	dBΩ	Less critical than max gain
Transimpedance Gain Step		0.25	0.5	dB	
Peak peak input current	125		400	uApp	Updated Max
Diff Output Swing, Peak-to-Peak	100		500	mVpp	
Input-Referred Noise (rms). Excludes PD shot noise.			2	uA_rms	Integrated from DC to 1.5xNyquist. Critical spec.
High-pass 3dB Bandwidth			100	kHz	Set by DCOC loop BW
Low-pass 3dB Bandwidth	50			GHz	
Return Loss (SDD22) @ Nyquist Freq			15	dB	
THD			3	%	At max swing
Group Delay Variation			3	ps	

Metric	Requirement			Unit	Comments
	Min	Typ	Max		
Data operating range (PAM4)	106.25		224	Gbps	
Power Consumption			0.2	pJ/bit	Power may be traded off to achieve key specs and schedule, but should not exceed 0.3 pJ/bit
Max Transimpedance Gain	66			dBΩ	Mid-band gain
Min. Transimpedance Gain			54	dBΩ	Less critical than max gain
Transimpedance Gain Step		0.25	0.5	dB	
CTLE peaking	0		3	dB	At Nyquist. Programmable in 1dB steps. Verification coverage may be relaxed to achieve schedule, with full gain-range verification at pre-layout, and post-layout verification at all gain settings at TT and at the minimum (0 dB) and maximum (3 dB) gain settings across SS and FF corners.
Max input DC current			520	uA	Correctable by a DC offset cancellation loop
Output Current	0		100	uA	1:1 mirror min/typ DC current N:1 divider max input DC current
Output Current Noise (referred to TIA input)		95		nA	RMS integrated from 5 kHz to 1 MHz @ 75 dBΩ, 80 uW optical input power
X-Dimension			0.4	mm	Tentative, to be confirmed
Y-Dimension			0.2	mm	Tentative, to be confirmed
Poly orientation		Y			Tentative, to be confirmed

Metric	Requirement			Unit	Comments
	Min	Typ	Max		
Data operating range (PAM4)	106.25		224	Gbps	
Power Consumption			0.2	pJ/bit	Power may be traded off to achieve key specs and schedule, but should not exceed 0.3 pJ/bit
Max Transimpedance Gain	66			dBΩ	Mid-band gain
Min. Transimpedance Gain			54	dBΩ	Less critical than max gain
Transimpedance Gain Step		0.25	0.5	dB	
Edge with high-speed signals		X edge			Tentative, to be confirmed

56-64G NRZ TIA

Electrical Requirements

Metric	Requirement			Unit	Comments
	Min	Typ	Max		
Data operating range (PAM4)	28		64	Gbps	
Power Consumption			0.2	pJ/bit	
Peak peak input current	50		400	uA	Updated Max
Max Transimpedance Gain	80	82		dBΩ	Mid-band gain
Min Transimpedance Gain			68	dBΩ	Mid-band gain
Transimpedance Gain Step			0.5	dB	
Diff Output Swing, Peak-to-Peak			600	mVpp	

Integrated Input-Referred Noise			1	uArms	Integrated from DC to 1.5xNyquist
High-pass 3dB Bandwidth			50	kHz	
Low-pass 3dB Bandwidth	30			GHz	
Return Loss (SDD22) @ Nyquist Freq			15	dB	
THD			8	%	At max swing
Group Delay Variation			5	pS	
Max input DC current			520	uA	Correctable by a DC offset cancellation loop
Output Current	0		100	uA	1:1 mirror min/typ DC current N:1 divider max input DC current
Output Current Noise (referred to TIA input)		95		nA	RMS integrated from 5 kHz to 1 MHz @ 75 dBΩ, 80 uW optical input power
X-Dimension			0.3	mm	Tentative, to be confirmed
Y-Dimension			0.15	mm	Tentative, to be confirmed
Poly orientation		Y			Tentative, to be confirmed
Edge with high-speed signals		X edge			Tentative, to be confirmed

Process, Voltage, and Temperature corners, sign-off requirements

- Process Corners: TT, SS, FF, FS, SF, FFA, and SSA for analog cells.
- Temperature: 0 deg. C to 125 deg. C
- Absolute Voltage Supplies: TBD
- Power Supply variation: +/- 5%